THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -14	Put $p(z) = 0$.
2	**(d)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
3	**(c)** -3	Put $y = 0$.
4	**(a)** 4	Substitute y and solve for x.
5	**(c)** $8x - 5$	Add like terms.
6	**(d)** 4	$p(2) - p(1) = a$.
7	**(c)** 82	Substitute n.
8	**(a)** parallel	Use the slope-intercept form.
9	**(b)** -30	Put $p(z) = 0$.
10	**(d)** 3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$m = \frac{13-(4)}{5-(2)}$ (1); $m = 3$ (1).	1+1
12	$p - q = 3x + 9$ (1); Zero $x = -3$ (1).	1+1
13	$a_1 = 5$, $a_2 = 13$, $a_3 = 21$ (1); $a_{10} = 77$ (1).	1+1
14	$m = \frac{8-(0)}{3-(-1)}$ (1); $m = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$18 = m(2) + 12$ (1); $m = 3$ (1); x-intercept: $0 = 3x + 12$, $x = -4$ (1).	1+1+1
16	Points $(0, 5)$, $(1, 1)$, $(2, -3)$ (1); Slope $= \frac{-3-(5)}{2-0}$ (1); $= -4$ (1).	1+1+1
17	$a_n = 7n - 1$ (1); $7n - 1 = 90$ (1); $n = 13$ (1).	1+1+1
	OR: $32 = m(3) + 20$ (1); $m = 4$ (1); x-intercept: $0 = 4x + 20$, $x = -5$ (1).	1+1+1
18	Points $(0, -6)$, $(1, 3)$, $(2, 12)$ (1); Slope $= \frac{12-(-6)}{2-0}$ (1); $= 9$ (1).	1+1+1
	OR: $a_n = 4n + 7$ (1); $4n + 7 = 75$ (1); $n = 17$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \frac{16-(8)}{3-(1)} = 4$ (1); $p(x) = 4x + 4$ (1); Zero: $x = -1$ (1); $p(5) = 24$ (2).	1+1+1+2
	OR: $E(n) = 150n + 5000$ (1); $E(20) = 8000$ (1); $150n + 5000 = 14000$, $n = 60$ (1); Extra items $= \frac{1200}{150} = 8$ (2).	1+1+1+2
20	Slope $= \frac{20-(5)}{6-(3)} = 5$ (1); $p(x) = 5x - 10$ (1); Zero: $x = 2$ (1); $p(4) = 10$ (2).	1+1+1+2
	OR: $E(n) = 200n + 6000$ (1); $E(15) = 9000$ (1); $200n + 6000 = 15000$, $n = 45$ (1); Extra items $= \frac{1200}{200} = 6$ (2).	1+1+1+2

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	Put $y = 0$.
2	**(b)** 6	Substitute y and solve for x.
3	**(a)** $6x - 5$	Add like terms.
4	**(a)** 9	$p(2) - p(1) = a$.
5	**(c)** 139	Substitute n.
6	**(c)** x-axis	Use the slope-intercept form.
7	**(a)** -27	Put $p(z) = 0$.
8	**(a)** -3	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
9	**(c)** -7	Put $y = 0$.
10	**(b)** 3	Substitute y and solve for x.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p - q = 4x + 8$ (1); Zero $x = -2$ (1).	1+1
12	$a_1 = 11$, $a_2 = 16$, $a_3 = 21$ (1); $a_{14} = 76$ (1).	1+1
13	$m = \frac{-1-(7)}{4-(0)}$ (1); $m = -2$ (1).	1+1
14	$p - q = 4x - 12$ (1); Zero $x = 3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$35 = m(4) + 15$ (1); $m = 5$ (1); x-intercept: $0 = 5x + 15$, $x = -3$ (1).	1+1+1
16	Points $(0, -2)$, $(1, -7)$, $(2, -12)$ (1); Slope $= \frac{-12-(-2)}{2-0}$ (1); $= -5$ (1).	1+1+1
17	$a_n = 9n - 6$ (1); $9n - 6 = 102$ (1); $n = 12$ (1).	1+1+1
	OR: $28 = m(5) + 18$ (1); $m = 2$ (1); x-intercept: $0 = 2x + 18$, $x = -9$ (1).	1+1+1
18	Points $(0, 3)$, $(1, 11)$, $(2, 19)$ (1); Slope $= \frac{19-(3)}{2-0}$ (1); $= 8$ (1).	1+1+1
	OR: $a_n = 6n + 8$ (1); $6n + 8 = 134$ (1); $n = 21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \frac{24-(12)}{4-(0)} = 3$ (1); $p(x) = 3x + 12$ (1); Zero: $x = -4$ (1); $p(6) = 30$ (2).	1+1+1+2
	OR: $E(n) = 120n + 4500$ (1); $E(25) = 7500$ (1); $120n + 4500 = 10500$, $n = 50$ (1); Extra items $= \frac{1080}{120} = 9$ (2).	1+1+1+2
20	Slope $= \frac{24-(6)}{5-(2)} = 6$ (1); $p(x) = 6x - 6$ (1); Zero: $x = 1$ (1); $p(3) = 12$ (2).	1+1+1+2
	OR: $E(n) = 250n + 7000$ (1); $E(12) = 10000$ (1); $250n + 7000 = 16000$, $n = 36$ (1); Extra items $= \frac{1750}{250} = 7$ (2).	1+1+1+2

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $5x + 11$	Add like terms.
2	**(d)** -3	$p(2) - p(1) = a$.
3	**(b)** 115	Substitute n.
4	**(d)** the origin	Use the slope-intercept form.
5	**(a)** -32	Put $p(z) = 0$.
6	**(d)** 4	Slope $= \frac{y_2 - y_1}{x_2 - x_1}$.
7	**(d)** -5	Put $y = 0$.
8	**(a)** 2	Substitute y and solve for x.
9	**(b)** $9x + 5$	Add like terms.
10	**(c)** 11	$p(2) - p(1) = a$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$a_1 = 5$, $a_2 = 14$, $a_3 = 23$ (1); $a_8 = 68$ (1).	1+1
12	$m = \frac{9-(-3)}{2-(-2)}$ (1); $m = 3$ (1).	1+1
13	$p - q = 5x - 10$ (1); Zero $x = 2$ (1).	1+1
14	$a_1 = 11$, $a_2 = 15$, $a_3 = 19$ (1); $a_{25} = 107$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$30 = m(1) + 24$ (1); $m = 6$ (1); x-intercept: $0 = 6x + 24$, $x = -4$ (1).	1+1+1
16	Points $(0, 7)$, $(1, 1)$, $(2, -5)$ (1); Slope $= \frac{-5-(7)}{2-0}$ (1); $= -6$ (1).	1+1+1
17	$a_n = 8n + 1$ (1); $8n + 1 = 121$ (1); $n = 15$ (1).	1+1+1
	OR: $35 = m(2) + 21$ (1); $m = 7$ (1); x-intercept: $0 = 7x + 21$, $x = -3$ (1).	1+1+1
18	Points $(0, -9)$, $(1, 1)$, $(2, 11)$ (1); Slope $= \frac{11-(-9)}{2-0}$ (1); $= 10$ (1).	1+1+1
	OR: $a_n = 3n + 17$ (1); $3n + 17 = 107$ (1); $n = 30$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Slope $= \frac{16-(8)}{3-(-1)} = 2$ (1); $p(x) = 2x + 10$ (1); Zero: $x = -5$ (1); $p(7) = 24$ (2).	1+1+1+2
	OR: $E(n) = 80n + 3000$ (1); $E(30) = 5400$ (1); $80n + 3000 = 9000$, $n = 75$ (1); Extra items $= \frac{800}{80} = 10$ (2).	1+1+1+2
20	Slope $= \frac{21-(7)}{6-(4)} = 7$ (1); $p(x) = 7x - 21$ (1); Zero: $x = 3$ (1); $p(2) = -7$ (2).	1+1+1+2
	OR: $E(n) = 300n + 8000$ (1); $E(10) = 11000$ (1); $300n + 8000 = 20000$, $n = 40$ (1); Extra items $= \frac{1500}{300} = 5$ (2).	1+1+1+2

Mistakes to avoid:

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- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
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